Which of these statements are correct?

(a) 1, 2 and 3 (b) 1 and 3

(c) 2 and 4 (d) 2, 3 and 4

Ans. (b) : Self-Sustainable Chain Reaction is possible in a nuclear reactor because each fission reaction has relatively more neutrons released and is slowed down by rapid neutron graphite.

81. Which one of the following is the correct sequence in increasing order of molecular weights of the hydrocarbons?

(a) Methane, ethane, propane and butane

(b) Propane, butane, ethane and methane

(c) Butane, ethane, propane and methane

(d) Butane, propane, ethane and methane

Ans. (a) : The molecular formulas and weights for given hydrocarbons can be written as:

Methane; CH_4, Molecular Weight; 16

Ethane; C_2H_6, Molecular weight; 30

Propane; C_3H_8, Molecular weight; 44

Butane:C_4H_{10}, Molecular weight: 58

Thus, the increasing order of molecular weights can be written as; Methane, ethane, propane and butane.

82. In an atom, the order of filling up of the orbitals is governed by:

(a) Aufbau principle

(b) Heisenberg's uncertainty principle

(c) Hund's rule

(d) Pauli's exclusion principle

Ans. (a) : Aufbau principle states that in the ground state of the atom, the orbitals are filled in order of their increasing energies, starting with the orbital of lowest energy. The word Aufbau is German word which means building up. The increasing order of energy and hence that of filling of orbitals is as follows: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p.

83. An aqueous solution of copper sulphate is acidic in nature because the salt undergoes:

(a) Dialysis (b) Electrolysis

(c) Hydrolysis (d) Photolysis

Ans. (c) : Aqueous solution of copper sulphate is acidic in nature because copper sulphate on dissolving with water, following chemical reaction takes place CuSO4(s) + 2H2O - Cu(OH)2(l) + H2SO4(aq) Now as we can see in above chemical equation the sulphuric acid generated is strong mineral acid which results into increase in acidity of solution. Thus aqueous solution of copper sulphate is acidic in nature.

84. Consider the following statements with reference to the Periodic Table of chemical elements:

1. Ionisation potential gradually decreases along a period.

2. In a group of elements, electron affinity decreases as the atomic weight increases.

3. In a given period, electronegativity decreases as the atomic number increases

Which of these statements(s) is/are correct?

(a) 1 only (b) 2 only

(c) 1 and 3 (d) 2 and 3

Ans. (b) : In a periodic table, in a group (vertical column) of element, electron affinity decreases as atomic weight increases. In a given period (horizontal row) electronegativity (tendency of atom to acquire negative charge) increases with increase in atomic number. Ionisation potential (the energy required to remove an electron) gradually increases along a period.

85. A radioactive substance has a half-life of four months. Three-fourth of the substance would decay in:

(a) 3 months (b) 4 months

(c) 8 months (d) 12 months

Ans. (c) : As it is given that half life of given substance is 4 months.

The amount of substance left after 4 months = 1/2

The amount of substance left after 8 months = 1/4

Therefore the amount of substance decay in 8 months = (1 – 1/4) = 3/4

86. Quartzite is metamorphosed from:

(a) Limestone (b) Obsidian

(c) Sandstone (d) Shale

Ans. (c) : Quartzite is metamorphosed from Sandstone. Quartzite is a nonfoliated metamorphic rock composed almost entirely of quartz. It forms when a quartz-rich sandstone is altered by the heat, pressure, and chemical activity of metamorphism. These conditions recrystallize the sand grains and the silica cement that binds them together.

87. Assertion (A) : A piece of copper and a piece of glass are heated to the same temperature. When touched, thereafter, the copper piece appears hotter than the glass piece.

Reason (R) : The density of cooper is more than that of glass.

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is NOT the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

Ans. (b) : Although the density of copper (8920–8960 Kg / m^3) is greater than the density of glass (2400–2800 Kg / m^3), its conductivity is due to the copper being warmer. Hence both the statement and the reason are true but the reason is not the correct explanation of the statement.

88. Assertion (A) : A chemical reaction becomes faster at higher temperatures.

Reason (R) : At higher temperatures, molecular motion becomes more rapid.

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is NOT the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

Ans. (a) : Temperature is the amount of kinetic energy possessed by the particles of an element and a reaction occurs when two atoms or particles collide with each other. For example, when an atom of Na collides with an atom of Cl, NaCl is formed. Now, when the temperature raises, the particles have more -kinetic energy and thus move faster. Moving faster implies that there will be more collisions per seconds Thus, the reaction occurs faster. Hence, Assertion and Reason both are correct and R is the correct explanation of A.

BIOLOGY

89. The American multinational company, Monsanto has produced an insect-resistant cotton variety that is undergoing field-trials in India. A toxin gene from which ONE of the following bacteria has been transferred to this transgenic cotton?

(a) Bacillus : substilis

(b) Bacillus : thuringiensis

(c) Bacillus : amyloliquijaciens

(d) Bacillus : globullii

Ans. (b) : The cotton incorporated the cry1Ac gene from the soil bacterium Bacillus thuringiensis (Bt), making the cotton toxic to bollworms.

90. The cellular and molecular control of programmed cell death is known as:

(a) Apoptosis (b) Ageing

(c) Degeneration (d) Necrosis

Ans. (a) : The action of apoptosis involves cellular and molecular control of programmed cell death. It occurs in multi cellular organism. Apoptosis is a highly regulated and controlled process that confers advantages during life cycle of an organism.

91. Match List I with List II and select the correct answer using the code given below the lists:

List I (Bone)	List II (Name)
A. Breast-bone	1. Clavicle
B. Collar-bone	2. Patella
C. Knee-cap	3. Scapula
D. Shoulder blade	4. Sternum

	A	B	C	D
(a)	4	1	3	2
(b)	1	4	3	2
(c)	1	4	2	3
(d)	4	1	2	3

Ans. (d) : Sternum is the breast bone. Clavicle is the collar bone. Patella is the knee cap. Scapula is the shoulder blade.

92. Consider the following statements:

1. Tapeworm is a hermaphrodite
2. Round-worm has separate sexes.
3. Filaria is caused by a nematode.
4. Guinea-worm is and annelid.

Which of these are correct?

(a) 1 and 2 (b) 1, 2 and 3

(c) 3 and 4 (d) 2, 3 and 4

Ans. (b) : Tape worms are hermaphrodite (both male and female reproductive organ present).

- Round worms are unisexual (separate sex).
- Filaria is caused by Wuchereria bancrofti which is a parasitic nematode.
- Guinea worm is a long and very thin nematode, not an annelid.

93. Which organelle in the Cell, other than nucleus, contains DNA?

(a) Centriole (b) Golgi apparatus

(c) Lysosome (d) Mitochondrion

Ans. (d) : Mitochondria contain DNA other than nucleus. Mitochondria are thought to be the parasites in the eukaryotic cells that got inside the latter as a result of the endo-symbiotic event that took place earlier in the evolution. Centriole is a small set of microtubules. Golgi apparatus is a complex of vesicles and folded membranes within the cytoplasm of most eukaryotic cells involved in secretion and intracellular transport. Lysosome is an organelle in the cytoplasm of eukaryotic cells containing degradative enzymes enclosed in a membrane.

94. "Athlete's Foot" is a disease caused by:

(a) Bacteria (b) Fungus

(c) Protozoan (d) Nematode

Ans. (b) : Athlete's Foot disease is caused by parasitic fungus of genus Trichophyton. Scaling, flaking and itching of affected areas are the symptoms of this disease. This disease transmitted in moist areas where people walk bare foot.

95. In the eye donation, which part of the eye is transplanted from the donor?

(a) Cornea (b) Lens

(c) Retina (d) The whole eye